

Vol 10 Chapter 2 — Complex Analysis and Holomorphic Functions

Keeper (author pass — deep math/physics revision)

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Chapter 2 — Complex Analysis and Holomorphic Functions

Level 1 — one sentence

Complex analysis — holomorphic functions, Cauchy's integral theorem, residue calculus, analytic continuation, conformal maps — is foundational for QFT (propagators, Wick rotation), special functions (Gamma, zeta), and the BST substrate's Bergman framework (holomorphic L^2 functions on bounded symmetric domains).

Level 2 — graduate-physicist precision

2.1 Holomorphic functions

A complex function $f : U \rightarrow \mathbb{C}$ is holomorphic if the complex derivative

$$f'(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}$$

exists at every $z \in U$. Equivalently: Cauchy-Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

for $f = u + iv$.

Holomorphic functions are infinitely differentiable (smooth) and locally given by power series.

2.2 Cauchy's integral theorem

For holomorphic f in a simply connected region U and closed curve $\gamma \subset U$:

$$\oint_{\gamma} f(z) dz = 0$$

Corollary (Cauchy integral formula): $f(z_0) = (2\pi i)^{-1} \oint_{\gamma} f(z)/(z - z_0) dz$ for z_0 inside γ .

2.3 Residue theorem

For meromorphic f with isolated poles at z_k inside γ :

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_k \text{Res}(f, z_k)$$

The residue at z_k is $\text{Res}(f, z_k) = \lim_{z \rightarrow z_k} (z - z_k) f(z)$ (simple pole).

Used extensively to evaluate real integrals via contour deformation.

2.4 Analytic continuation

If f is holomorphic on U_1 and g on U_2 with $U_1 \cap U_2 \neq \emptyset$ and $f = g$ on $U_1 \cap U_2$, then f and g are the same function on $U_1 \cup U_2$. Holomorphic functions have unique analytic continuations.

The Riemann zeta function $\zeta(s) = \sum n^{-s}$ converges for $\Re s > 1$ and analytically continues to $\mathbb{C} \setminus \{1\}$. Hardy's $\zeta(0) = -1/2$, $\zeta(-1) = -1/12$ are analytic continuations.

2.5 Conformal maps

Conformal map = holomorphic and locally angle-preserving. Möbius transformations $z \rightarrow (az + b)/(cz + d)$ are the conformal maps of the Riemann sphere.

Riemann mapping theorem: any simply connected domain $\neq \mathbb{C}$ is conformally equivalent to the unit disk.

2.6 Bergman holomorphic L^2 on D_{IV}^5

The Bergman space $H^2(D_{IV}^5)$ consists of L^2 -integrable holomorphic functions on the bounded symmetric domain D_{IV}^5 . This is the BST substrate Hilbert space (Vol 5 Ch 1, Vol 11 Ch 2).

The Bergman kernel $K(z, w)$ is the reproducing kernel: $f(w) = \int_{D_{IV}^5} f(z) \overline{K(z, w)} d\mu(z)$.

Substrate-mechanism reading: the substrate's natural function class on D_{IV}^5 is holomorphic L^2 , with the kernel encoding the substrate's Zone 3 commitment projection (Vol 5 Ch 7).

2.7 K-audit anchors

- Vol 5 Ch 1: Bergman H^2 as substrate Hilbert space
- Vol 11 Ch 2: Bergman kernel (foundational)

Level 3 — 5th-grader accessibility

Complex analysis: calculus of complex-valued functions. Holomorphic = complex-differentiable. Holomorphic functions are super-smooth — knowing them on a tiny region determines them everywhere by analytic continuation. Cauchy's theorem: closed-contour integrals of holomorphic functions are zero. Residue theorem: contour integrals around poles = $2\pi i \times$ sum of residues. Tons of applications: real integrals evaluated via contour deformation, zeta-function values, propagators in QFT. Bergman space of holomorphic L^2 functions on a domain — the substrate's natural function class in BST.

What comes next

Chapter 3 develops ODEs and Sturm-Liouville theory.

Where to look this up

- Ahlfors, *Complex Analysis*
- BST: Vol 5 Ch 1, Vol 11 Ch 2